MikroTik RouterOS
VLAN Creation & IP Assignment

Hands-on practice walkthrough (WebFig / Winbox-style GUI + CLI equivalents)

Item	Value
Platform	MikroTik router, RouterOS 7.x (GUI shows "RouterOS 7.25 available")
Tasks covered	1) Create a VLAN interface 2) Assign an IP address to that VLAN
Parent interface	10G-SFP+
VLAN name / ID	indadul-FNA / 3031
IP assigned	11.10.10.2/30
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Flow: Interfaces → VLAN → New → set Name / VLAN ID / Interface → OK. Then IP → Addresses → New → set Address / Interface → OK.

1. Concepts in 60 seconds

- **VLAN (802.1Q):** a tag (ID 1-4094) added to Ethernet frames so one physical link carries multiple isolated Layer-2 networks.
- **VLAN interface (RouterOS):** a logical interface bound to a parent interface (here `10G-SFP+`). It sends/receives only frames tagged with its VLAN ID.
- **IP on a VLAN:** the router needs an IP on the VLAN interface to route/talk Layer-3 on that VLAN. The VLAN ID must match the far end (switch/OLT/upstream router).
- **/30 subnet:** 4 addresses = network, 2 usable hosts, broadcast. Standard for point-to-point links.

Field	Value for 11.10.10.2/30
Network	11.10.10.0
Usable hosts	11.10.10.1 and 11.10.10.2
Broadcast	11.10.10.3
Netmask	255.255.255.252
Peer (expected)	11.10.10.1 (the other end of the link)

2. Part 1 - Create the VLAN interface

Step 1: Open the VLAN tab and click New

Left menu: **Interfaces** → tab **VLAN** → **New**. The list shows existing VLAN interfaces (name, MTU, VLAN ID, parent interface, Tx/Rx).

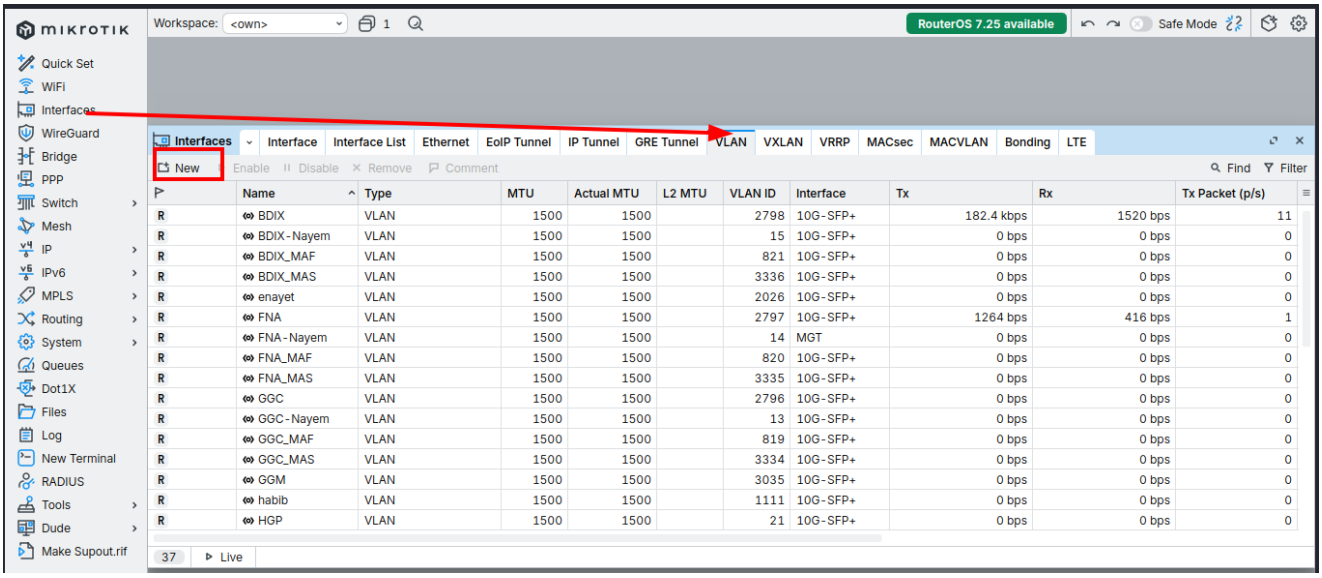


Figure 1 - Interfaces → VLAN tab → New (arrow = path, red box = New button)

Step 2: Fill in the VLAN parameters

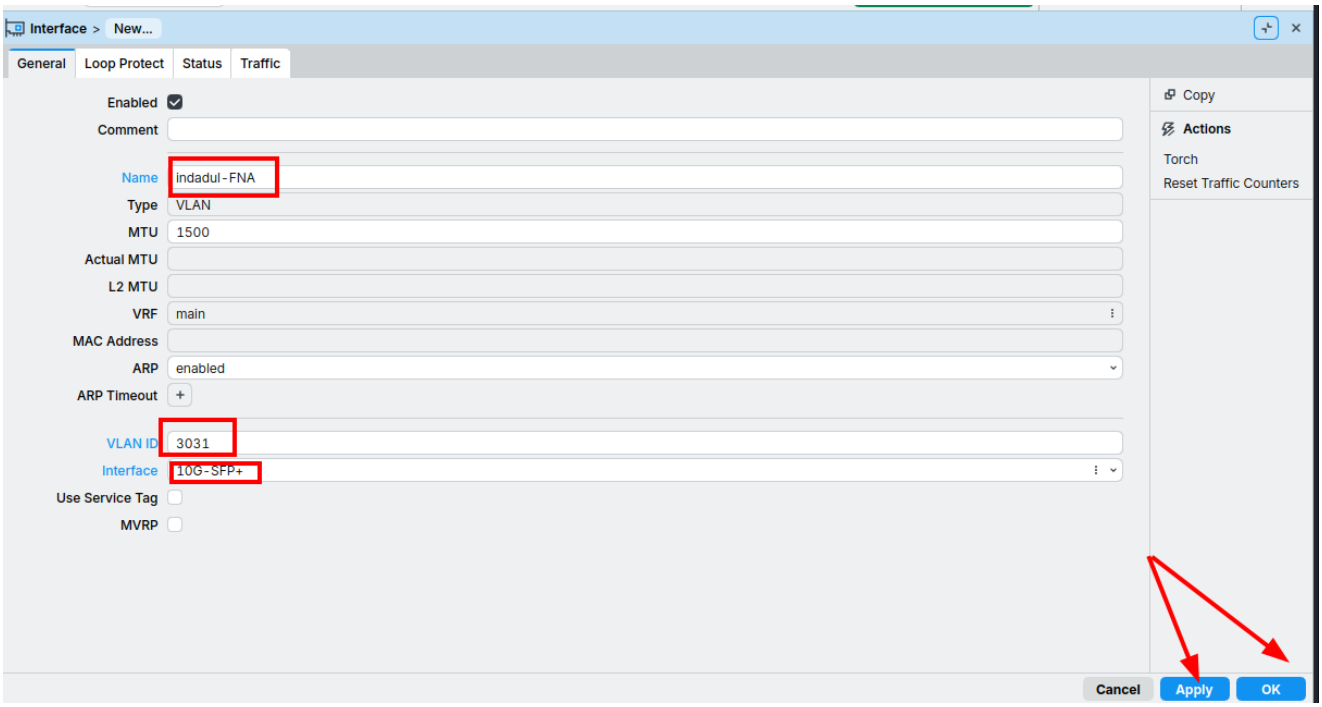


Figure 2 - New VLAN interface dialog (Name, VLAN ID, Interface highlighted)

Field	Value used	Notes
Name	indadul-FNA	Any unique label. Keep a clear naming convention (service-site).
VLAN ID	3031	Must match the tag configured on the other end of the link.
Interface	10G-SFP+	Parent/physical interface carrying the tagged traffic.
MTU	1500	Default. Raise parent MTU first if you need larger frames (e.g. QinQ).
Use Service Tag	unchecked	Only for 802.1ad (QinQ) S-tag.
ARP	enabled	Default.

Step 3: Apply / OK

Click **Apply** to save and stay on the dialog, or **OK** to save and close. The new interface appears in the VLAN list with the **R** (running) flag if the parent link is up.

CLI equivalent

```
/interface vlan add name=indadul-FNA vlan-id=3031 interface=10G-SFP+ comment="practice"
/interface vlan print where name=indadul-FNA
```

3. Part 2 - Assign an IP address to the VLAN

Step 4: Open IP → Addresses → New

Left menu: **IP** → **Addresses** → **New**. The list shows each address with its network and the interface it is bound to (one row per VLAN interface). Addresses were blurred in the screenshot for privacy.

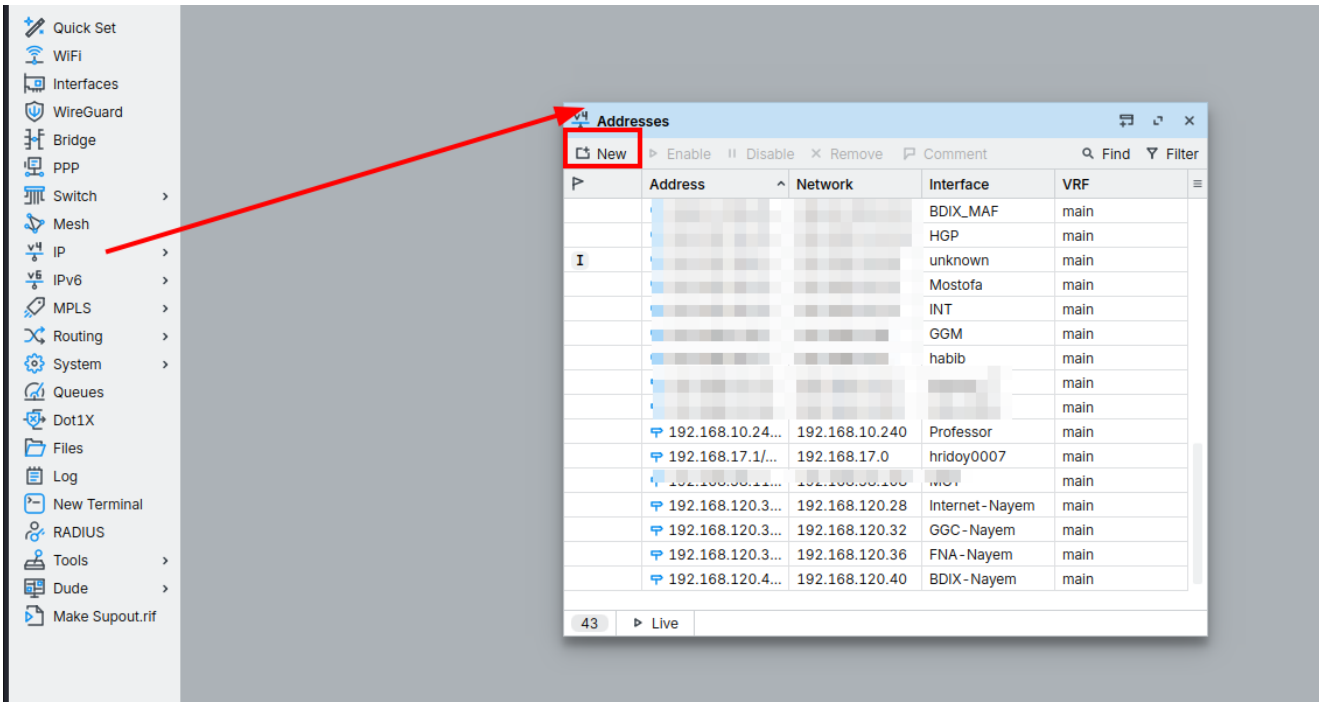


Figure 3 - IP → Addresses window, New button highlighted

Step 5: Set address and interface

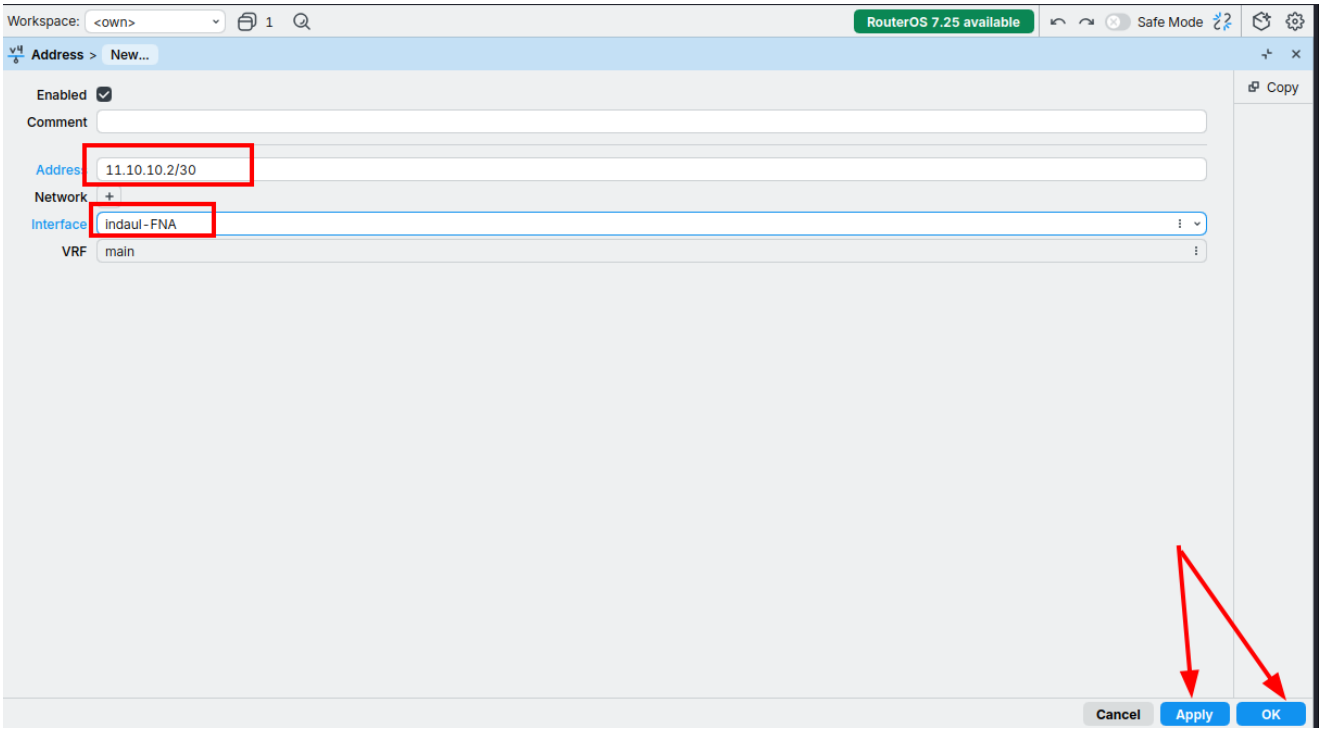


Figure 4 - New Address dialog: Address 11.10.10.2/30, Interface = the VLAN interface

Field	Value used	Notes
Address	11.10.10.2/30	IP and prefix length together. Use CIDR format.
Network	(auto)	RouterOS calculates 11.10.10.0 automatically.
Interface	indadul-FNA	Pick the VLAN interface, not the physical port.
VRF	main	Default routing table/VRF.

Step 6: Apply / OK

Click **Apply** then **OK**. A connected route for `11.10.10.0/30` is created automatically on `indadul-FNA`.

CLI equivalent

```
/ip address add address=11.10.10.2/30 interface=indadul-FNA comment="practice"
/ip address print where interface=indadul-FNA
```

4. Verification commands

```
# VLAN exists and is running
/interface vlan print detail where name=indadul-FNA

# IP and connected route
/ip address print where interface=indadul-FNA
/ip route print where dst-address=11.10.10.0/30

# Reachability to the peer (other end of the /30)
/ping 11.10.10.1 count=4

# ARP entry for the peer
/ip arp print where interface=indadul-FNA

# Live traffic and tag inspection on the VLAN
/interface monitor-traffic indadul-FNA
/tool torch interface=indadul-FNA

# Parent link state
/interface ethernet print where name=10G-SFP+
```

Check	Good result
VLAN flag	R (running) in /interface vlan print
Address flag	No X (disabled) or I (invalid) flag
Ping peer	Replies from 11.10.10.1 (requires peer configured with same VLAN ID and 11.10.10.1/30)
Torch	Packets seen with VLAN 3031 on the parent; traffic on the VLAN interface

5. Notes, pitfalls, and best practices

- **Name consistency:** in Figure 2 the VLAN is named `indadul-FNA`, but in Figure 4 the interface reads `indaul-FNA`. The Interface field is a dropdown of existing interfaces, so confirm the exact name with `/interface vlan print` and re-select it if they differ.
- **VLAN ID must match the far end.** Wrong ID = no ping, even if everything on your router looks fine.
- **Bind IP to the VLAN interface, not the physical port** (10G-SFP+) unless you want untagged traffic there.
- **Addressing:** 11.10.10.0/30 is public (non-RFC1918) space. For labs and practice prefer private ranges such as 10.x.x.x, 172.16-31.x.x or 192.168.x.x, or use the real assigned block only if it is yours/provided by your upstream.
- **Overlap:** RouterOS lets you add overlapping subnets on different interfaces; avoid it, it breaks routing. Check `/ip address print` first.
- **If the parent is a bridge port:** with bridge VLAN filtering enabled you must also configure `/interface bridge vlan`; plain VLAN interfaces on a physical port as shown here need no bridge.
- **Safe Mode** (top bar toggle) auto-reverts changes if you lose connection. Use it when editing remote routers.
- **Backup before changes:** `/system backup save name=pre-vlan` or `/export file=pre-vlan`.
- **Privacy:** blur customer/ISP IPs and VLAN names before posting screenshots (as done in Figure 3).

6. Quick reference - full CLI script

```
# Create VLAN
/interface vlan add name=indadul-FNA vlan-id=3031 interface=10G-SFP+

# Assign IP
/ip address add address=11.10.10.2/30 interface=indadul-FNA

# Test
/ping 11.10.10.1 count=4

# Undo
/ip address remove [find interface=indadul-FNA]
/interface vlan remove [find name=indadul-FNA]
```

GUI path	CLI path
Interfaces > VLAN	/interface vlan
IP > Addresses	/ip address
IP > Routes	/ip route
Tools > Torch	/tool torch
Tools > Ping	/ping

7. Next topics to practice

- Static routes and default gateway (`/ip route`)
- DHCP server on a VLAN
- Bridge VLAN filtering / trunk and access ports
- NAT (masquerade) and basic firewall filter rules
- Simple queues for per-VLAN bandwidth limiting
- OSPF/BGP basics for ISP-style peering